



UNIVERSITÀ DEGLI STUDI DI PISA

CORSO DI STUDIO IN BIONICS ENGINEERING

STATISTICAL SIGNAL PROCESSING - *Prova scritta del 15/02/2024*

PROBLEM - We observe N samples of the following discrete-time real-valued random process:

$$X[n] = S[n] + W[n] = A \cos(2\pi f_0 n) + W[n], \quad n = 0, 1, 2, \dots, N-1,$$

where A is a constant signal modelled as a Gaussian distributed random variable:

$$f_A(a) = \frac{1}{\sqrt{2\pi\sigma_A^2}} e^{-\frac{a^2}{2\sigma_A^2}}, \quad \text{where } E\{A\} = 0, \quad E\{A^2\} = \sigma_A^2,$$

f_0 is a-priori known, and $W[n]$ is additive white Gaussian noise (AWGN), with zero mean and power σ_w^2 . Define γ as the signal to noise power ratio (SNR), i.e. $\gamma = E\{\|\mathbf{S}\|^2\} / E\{\|\mathbf{W}\|^2\}$, where $\mathbf{S}$ and $\mathbf{W}$ are the signal and noise random vectors, defined as $\mathbf{S} = [S[0] \ S[1] \ \dots \ S[N-1]]^T$ and $\mathbf{W} = [W[0] \ W[1] \ \dots \ W[N-1]]^T$.

Derive:

- (1) the expression of γ , the mean function $\eta_X[n] = E\{X[n]\}$, and the autocorrelation function (ACF) $R_X[n, m] = E\{X[n]X[n+m]\}$ of the observed process $X[n]$; establish if the process is wide sense stationary (w.s.s.) or not;
- (2) the mean vector $\mathbf{m}_X$ and the covariance matrix $\mathbf{C}_X$ of the observed data vector: $\mathbf{X} = \mathbf{S} + \mathbf{W}$;
- (3) the Maximum A Posteriori (MAP) estimator of the random parameter A given the observation of $\mathbf{X}$ and express it as a function of γ , of the sample size N , and of the Maximum Likelihood (ML) estimator of A (which in this case is the well-known matched filter output); discuss what happens to $\hat{A}_{MAP}$ when $N \rightarrow \infty$ and when $\gamma \rightarrow \infty$ (keeping constant σ_w^2);
- (4) the bias and the mean square error (MSE) of the MAP estimator of A and express it as a function of γ , of the sample size N , and of the noise power σ_w^2 ; discuss what happens to $MSE\{\hat{A}_{MAP}\}$ when $N \rightarrow \infty$ and when $\gamma \rightarrow \infty$ (keeping constant σ_w^2); plot $MSE\{\hat{A}_{MAP}\}$ as a function of γ and as a function of the sample size N .
- (5) the Bayesian Cramér-Rao bound (BCRB) on the estimate of A and determine whether the MAP estimator is efficient or not.



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Solution

(1) SNR, Mean function and ACF of the process $X[n]$.

$$\gamma = \frac{E\{\|\mathbf{S}\|^2\}}{E\{\|\mathbf{W}\|^2\}} = \frac{E\left\{\sum_{n=0}^{N-1} S^2[n]\right\}}{E\left\{\sum_{n=0}^{N-1} W^2[n]\right\}} = \frac{E\{A^2\} \sum_{n=0}^{N-1} \cos^2(2\pi f_0 n)}{\sum_{n=0}^{N-1} E\{W^2[n]\}} = \frac{\sigma_A^2 \frac{N}{2}}{\sigma_W^2 N} = \frac{\sigma_A^2}{2\sigma_W^2}$$

$$\eta_X[n] = E\{X[n]\} = E\{A\} \cos(2\pi f_0 n) + E\{W[n]\} = 0$$

$$\begin{aligned} R_X[n, m] &= E\{X[n]X[n+m]\} = E\{(A \cos(2\pi f_0 n) + W[n])(A \cos(2\pi f_0(n+m)) + W[n+m])\} \\ &= E\{A^2\} \cos(2\pi f_0 n) \cos(2\pi f_0(n+m)) + E\{W[n]W[n+m]\} \\ &= \frac{E\{A^2\}}{2} \cos(2\pi f_0 m) + \frac{E\{A^2\}}{2} \cos(2\pi f_0(2n+m)) + R_W[m] \\ &= \frac{\sigma_A^2}{2} \cos(2\pi f_0 m) + \frac{\sigma_A^2}{2} \cos(2\pi f_0(2n+m)) + \sigma_W^2 \delta[m] \end{aligned}$$

Since the ACF depends both on m and n , the process is **not** w.s.s.

(2) Mean vector and covariance matrix of the data vector $\mathbf{X}$.

$$\mathbf{m}_X = E\{\mathbf{X}\} = E\{A\mathbf{c} + \mathbf{W}\} = E\{A\}\mathbf{c} + E\{\mathbf{W}\} = \mathbf{0}$$

$$\text{where } \mathbf{c} = [1 \quad \cos(2\pi f_0) \quad \cdots \quad \cos(2\pi f_0(N-1))]^T$$

$$\text{Note that } \|\mathbf{c}\|^2 = \mathbf{c}^T \mathbf{c} = \sum_{n=0}^{N-1} \cos^2(2\pi f_0 n) = \frac{N}{2}$$

$$\mathbf{C}_X = E\{\mathbf{X}\mathbf{X}^T\} = E\{(A\mathbf{c} + \mathbf{W})(A\mathbf{c} + \mathbf{W})^T\} = E\{A^2\}\mathbf{c}\mathbf{c}^T + \mathbf{C}_W = \sigma_A^2 \mathbf{c}\mathbf{c}^T + \sigma_W^2 \mathbf{I}$$

(3) Maximum A Posteriori (MAP) estimator of A

$$\hat{A}_{MAP} = \arg \max_a f_{A|X}(a|\mathbf{x}) = \arg \max_a f_X(\mathbf{x}, a) = \arg \max_a \ln f_X(\mathbf{x}, a)$$



$$f_X(\mathbf{x}, a) = f_X(\mathbf{x}|a)f_A(a) = f_X(\mathbf{x}|a) \cdot \frac{1}{\sqrt{2\pi\sigma_A^2}} e^{-\frac{a^2}{2\sigma_A^2}}$$

The conditional PDF is Gaussian: $\mathbf{x}|A \in N(\mathbf{Ac}, \sigma_w^2 \mathbf{I})$. Taking into account that the noise samples are IID, the conditional pdf (in scalar and vector notation) is given by:

$$\begin{aligned} f_X(\mathbf{x}|a) &= \prod_{n=0}^{N-1} \frac{1}{\sqrt{2\pi\sigma_w^2}} \exp\left(-\frac{(X[n]-a)^2}{2\sigma_w^2}\right) = \frac{1}{\sqrt{(2\pi)^N |\sigma_w^2 \mathbf{I}|}} \exp\left(-\frac{1}{2}(\mathbf{x}-\mathbf{ac})^T (\sigma_w^2 \mathbf{I})^{-1} (\mathbf{x}-\mathbf{ac})\right) \\ &= \frac{1}{\left(\sqrt{2\pi\sigma_w^2}\right)^N} \exp\left(-\frac{1}{2\sigma_w^2}(\mathbf{x}-\mathbf{ac})^T (\mathbf{x}-\mathbf{ac})\right) \end{aligned}$$

Hence, the logarithm of the joint pdf of $\mathbf{X}$ and A can be expressed as:

$$\ln f_X(\mathbf{x}, a) = -\frac{N}{2} \ln(2\pi\sigma_w^2) - \frac{1}{2\sigma_w^2}(\mathbf{x}-\mathbf{ac})^T (\mathbf{x}-\mathbf{ac}) - \frac{1}{2} \ln(2\pi\sigma_A^2) - \frac{a^2}{2\sigma_A^2}$$

that can be rearranged as:

$$\begin{aligned} \ln f_X(\mathbf{x}, a) &= k - \frac{1}{2\sigma_w^2}(\mathbf{x}^T \mathbf{x} - \mathbf{ac}^T \mathbf{x} - \mathbf{a} \mathbf{x}^T \mathbf{c} + a^2 \mathbf{c}^T \mathbf{c}) - \frac{a^2}{2\sigma_A^2} = -\left(\frac{N}{4\sigma_w^2} + \frac{1}{2\sigma_A^2}\right)a^2 + \left(\frac{1}{\sigma_w^2} \mathbf{c}^T \mathbf{x}\right)a + k \\ &= -\frac{N}{4\sigma_w^2} \left(1 + \frac{2\sigma_w^2}{N\sigma_A^2}\right)a^2 + \frac{N}{2\sigma_w^2} \left(\frac{2}{N} \mathbf{c}^T \mathbf{x}\right)a + k = -\frac{N}{4\sigma_w^2} \left(1 + \frac{1}{N\gamma}\right)a^2 + \frac{N}{2\sigma_w^2} \left(\frac{2}{N} \mathbf{c}^T \mathbf{x}\right)a + k \end{aligned}$$

where k includes all terms that do not depend on a , and so are irrelevant.

The MAP estimator is obtained by solving the MAP equation:

$$\frac{d \ln f_X(\mathbf{x}, a)}{da} = -\frac{N}{2\sigma_w^2} \left(1 + \frac{1}{N\gamma}\right)a + \frac{N}{2\sigma_w^2} \left(\frac{2}{N} \mathbf{c}^T \mathbf{x}\right) = 0$$

The solution of this 1st order linear equation is:

$$a = \frac{1}{\left(1 + \frac{1}{N\gamma}\right)} \left(\frac{2}{N} \mathbf{c}^T \mathbf{x}\right) = \frac{N\gamma}{1 + N\gamma} \left(\frac{2}{N} \mathbf{c}^T \mathbf{x}\right) = \alpha \hat{A}_{ML} + (1 - \alpha) \eta_A = \alpha \hat{A}_{ML}$$



where $\hat{A}_{ML} = \frac{2}{N} \mathbf{c}^T \mathbf{x} = \frac{2}{N} \sum_{n=0}^{N-1} X[n] \cos(2\pi f_0 n)$ and $\eta_A = 0$, and we defined $\alpha \triangleq \frac{N\gamma}{1+N\gamma}$.

When $N \rightarrow \infty$ or $\gamma \rightarrow \infty$ (i.e. $\sigma_A^2 \rightarrow \infty$) we have: $\alpha \rightarrow 1$ so that $\hat{A}_{MAP} \rightarrow \hat{A}_{ML}$.

When $N \rightarrow 0$ or $\gamma \rightarrow 0$ (i.e. $\sigma_A^2 \rightarrow 0$) we have: $\alpha \rightarrow 0$ so that $\hat{A}_{MAP} \rightarrow 0$, which is the a-priori mean of A .

(4) Bias and MSE of the MAP estimator of A

$$\hat{A}_{MAP} = \alpha \hat{A}_{ML}$$

$$\text{bias}\{\hat{A}_{MAP}\} = E\{A - \hat{A}_{MAP}\} = E\{A\} - \alpha E\{\hat{A}_{ML}\} = -\alpha E\{A\} = 0, \text{ since } \eta_A = 0.$$

Hence, the MAP estimator is **unbiased**.

$$\begin{aligned} \text{MSE}\{\hat{A}_{MAP}\} &= E\left\{\left(A - \hat{A}_{MAP}\right)^2\right\} = E\left\{\left(A - \alpha \hat{A}_{ML}\right)^2\right\} \\ &= E\left\{\left(A - \alpha A + \alpha A - \alpha \hat{A}_{ML}\right)^2\right\} = E\left\{\left[\alpha(A - \hat{A}_{ML}) + (1-\alpha)A\right]^2\right\} \\ &= \alpha^2 E\left\{\left(A - \hat{A}_{ML}\right)^2\right\} + (1-\alpha)^2 E\{A^2\} + 2\alpha(1-\alpha)E\left\{\left(A - \hat{A}_{ML}\right)A\right\} \\ &= \alpha^2 \frac{2\sigma_w^2}{N} + (1-\alpha)^2 \sigma_A^2 + 2\alpha(1-\alpha)E\left\{\frac{2}{N} \mathbf{c}^T \mathbf{W} A\right\} = \alpha^2 \frac{2\sigma_w^2}{N} + (1-\alpha)^2 \sigma_A^2 \\ &= \frac{2\sigma_w^2}{N} \left(\alpha^2 + (1-\alpha)^2 N\gamma\right) = \frac{2\sigma_w^2}{N} \left(\frac{(N\gamma)^2}{(N\gamma+1)^2} + \frac{N\gamma}{(N\gamma+1)^2}\right) \\ &= \frac{2\sigma_w^2}{N} \left(\frac{N\gamma}{N\gamma+1}\right) = \frac{2\sigma_w^2 \gamma}{N\gamma+1} = \frac{\sigma_A^2}{N\gamma+1} \end{aligned}$$

Moreover, we observe that $\text{MSE}\{\hat{A}_{MAP}\} = \frac{2\sigma_w^2}{N} \left(\frac{N\gamma}{N\gamma+1}\right) = \alpha \text{MSE}\{\hat{A}_{ML}\}$, since it is well-known

that for this estimation problem we have $\text{MSE}\{\hat{A}_{ML}\} = \frac{2\sigma_w^2}{N}$.



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When $N \rightarrow \infty$ or $\gamma \rightarrow \infty$ (i.e. $\sigma_A^2 \rightarrow \infty$) we have: $\alpha \rightarrow 1$ so $MSE\{\hat{A}_{MAP}\} \rightarrow MSE\{\hat{A}_{ML}\} = \frac{2\sigma_w^2}{N}$,

which tends to 0 when $N \rightarrow \infty$, but not when $\gamma \rightarrow \infty$ if $\sigma_w^2 = const \neq 0$.

When $N \rightarrow 0$ or $\gamma \rightarrow 0$ (i.e. $\sigma_A^2 \rightarrow 0$) we have: $\alpha \rightarrow 0$ so that $\hat{A}_{MAP} \rightarrow 0$. However, we have to make a distinction here:

When $N \rightarrow 0$ and $\gamma \neq 0$, i.e. $\sigma_A^2 \neq 0$, $\hat{A}_{MAP} \rightarrow 0 = \eta_A$ and $MSE\{\hat{A}_{MAP}\} \rightarrow \frac{2\sigma_w^2\gamma}{N\gamma+1}\bigg|_{N=0} = 2\sigma_w^2\gamma = \sigma_A^2$.

When $\gamma \rightarrow 0$, i.e. $\sigma_A^2 \rightarrow 0$, and $N \neq 0$, we have $\hat{A}_{MAP} \rightarrow 0 = A$, so $MSE\{\hat{A}_{MAP}\} \rightarrow 0$. In fact, in the case that $\sigma_A^2 = 0$ we have that A is deterministic, equal to 0, and so it is perfectly “estimated”.

(5) Bayes Cramér-Rao bound (BCRB)

$$BCRB(A) = -\frac{1}{E\left\{\frac{d^2 \ln f_X(\mathbf{x}, a)}{da^2}\right\}}$$

$$\frac{d \ln f_X(\mathbf{x}, a)}{da} = -\frac{N}{2\sigma_w^2} \left(1 + \frac{1}{N\gamma}\right) a + \frac{N}{2\sigma_w^2} \left(\frac{2}{N} \mathbf{c}^T \mathbf{x}\right), \quad \frac{d^2 \ln f_X(\mathbf{x}, a)}{da^2} = -\frac{N}{2\sigma_w^2} \left(1 + \frac{1}{N\gamma}\right)$$

$$-E\left\{\frac{d^2 \ln f_X(\mathbf{x}, a)}{da^2}\right\} = E\left\{\frac{N}{2\sigma_w^2} \left(1 + \frac{1}{N\gamma}\right)\right\} = \frac{N}{2\sigma_w^2} \left(1 + \frac{1}{N\gamma}\right)$$

$$BCRB(A) = -\frac{1}{-E\left\{\frac{d^2 \ln f_X(\mathbf{x}, a)}{da^2}\right\}} = \frac{1}{\frac{N}{2\sigma_w^2} \left(1 + \frac{1}{N\gamma}\right)} = \frac{N\gamma}{1 + N\gamma} \cdot \frac{2\sigma_w^2}{N} = \alpha \frac{2\sigma_w^2}{N} = MSE\{\hat{A}_{MAP}\}$$

Hence, the MAP estimator is **efficient**.